

input image $(32 \times 32 \times 3)$ = number of patches/tokens per image $(32 \times 32) / (4 \times 4) = 64$ patches

patch size (4×4)

patch embedding

embedding dimension: 64

(hidden-dim: 128)

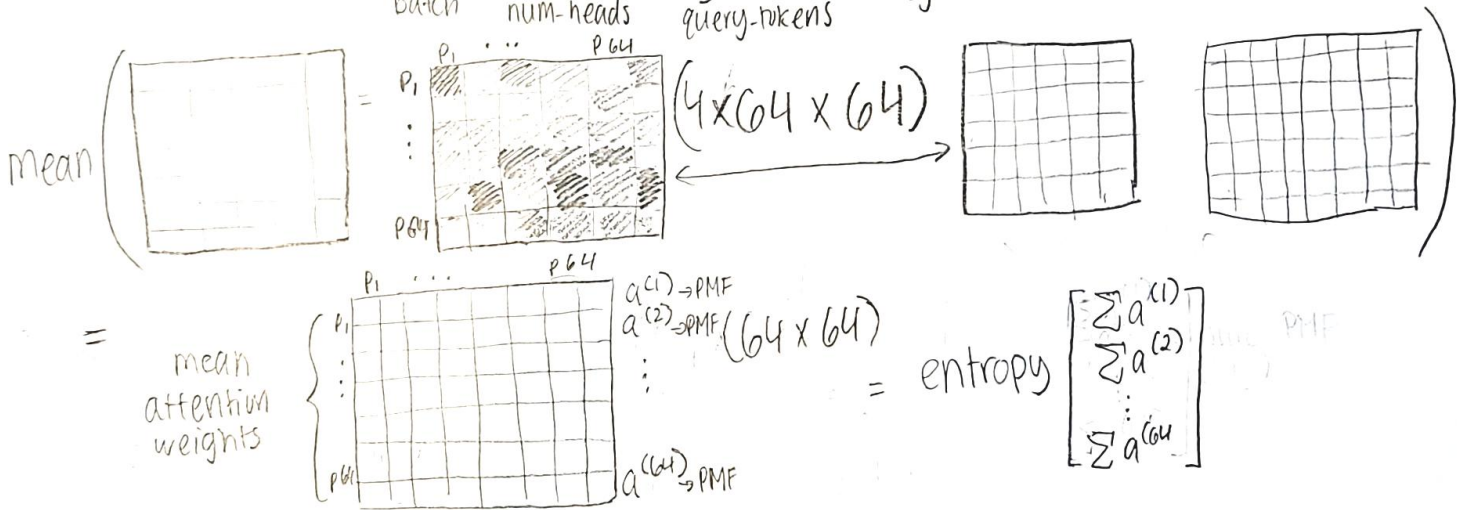
after Patch Embedding: $[1, 64, 64]$

batch num-patches embed-dim

Self attention on each token/patch

attention weights = $[1, 4, 64, 64]$ = $\text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{\text{hidden-dim}}}\right) \cdot V$

batch num-heads query-tokens key-tokens



interpreting attention weights as probability distribution over input token/patches

→ each one is a separate distribution

high entropy

= morespread of attention across the image for that patch

low entropy

= focused attention in regions of the image for that patch